REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 12 Advanced | Module 22 Lesson 4 | The Quantum Model of the Atom

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Photon transitions: $E = hf = hc/\lambda$. Matter-wave relation $\lambda = h/p$. A standing-wave illustration has $2\pi r = n\lambda$. $h = 6.626 \times 10^{-34}$ J s; $c = 3.00 \times 10^{8}$ m/s.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 12 book, printed pages 75–79.

P01. An electron cloud represents:

A. A spatial probability distribution B. A fixed classical orbit with exact position
C. A literal cloud of negatively charged gas D. Only the atomic nucleus

Answer

A. A spatial probability distribution.

P02. Stimulated emission occurs when an incident photon:

A. Always ionises the whole atom B. Triggers an excited system to emit a matching photon
C. Is necessarily absorbed without re-emission D. Destroys energy

Answer

B. Triggers an excited system to emit a matching photon.

P03. Laser light is typically:

A. Necessarily white and incoherent B. Without frequency
C. Highly coherent and directional D. Composed of electrons rather than photons

Answer

C. Highly coherent and directional.

P04. Population inversion means:

A. All particles have left the material B. The nucleus becomes negative
C. The atom has zero energy D. More particles occupy the relevant upper laser level than the lower one

Answer

D. More particles occupy the relevant upper laser level than the lower one.

P05. The quantum model describes an electron's location using:

A. Probabilities rather than an exact classical path B. A precisely known circular path at every instant
C. Only the atom's colour D. An arbitrary position unrelated to the state

Answer

A. Probabilities rather than an exact classical path.

P06. A laser emits photons of wavelength 640 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

A. 3.11×10^{-19} J B. 6.21×10^{-19} J
C. 1.55×10^{-19} J D. -3.11×10^{-19} J

Answer

A. 3.11×10^{-19} J. $E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(640 \times 10^{-9}) = 3.11 \times 10^{-19}$ J.

P07. A laser pulse contains 0.004 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

A. 2.67×10^{16} photons B. 1.33×10^{16} photons
C. 6.67×10^{15} photons D. -1.33×10^{16} photons

Answer

B. 1.33×10^{16} photons. $N = E_{pulse}/E_{photon} = 0.004/(3.00 \times 10^{-19}) = 1.33 \times 10^{16}$ photons.

P08. In a standing-wave picture, a circular path contains 3 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

A. 1.2 nm B. 0.3 nm
C. 0.6 nm D. -0.6 nm

Answer

C. 0.6 nm. $2\pi r = n\lambda$; circumference $= 3 \times 0.200 = 0.6$ nm.

P09. Find the energy of a photon of frequency 2×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

A. 2.65×10^{-19} J B. 6.63×10^{-20} J
C. -1.33×10^{-19} J D. 1.33×10^{-19} J

Answer

D. 1.33×10^{-19} J. $E = hf = (6.626 \times 10^{-34})(2 \times 10^{14}) = 1.33 \times 10^{-19}$ J.

P10. An atomic transition emits a photon of energy 2.5 eV. Calculate its wavelength in nanometres using $hc \approx 1240$ eV nm.

A. 496 nm B. 992 nm
C. 248 nm D. -496 nm

Answer

A. 496 nm. $\lambda = hc/E = 1240/2.5 = 496$ nm.

P11. What is an orbital in the quantum model?

Answer

A quantum state associated with an electron probability distribution, not a classical track followed by the electron.

P12. Distinguish spontaneous from stimulated emission.

Answer

Spontaneous emission occurs without an incident triggering photon; stimulated emission is triggered by suitable light and produces an additional coherent photon.

P13. State two typical properties of laser light.

Answer

It is highly directional and coherent; many lasers also have a narrow wavelength range.

P14. Why is energy supplied to a laser medium?

Answer

Pumping raises particles into excited states and helps establish conditions such as population inversion needed for net light amplification.

P15. Name one useful laser application and its physical purpose.

Answer

For example, optical communication uses controlled laser light to carry signals through fibres, or precision cutting uses concentrated light energy.

P16. A laser emits photons of wavelength 660 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

Answer

$E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(660 \times 10^{-9}) = 3.01 \times 10^{-19}$ J.

P17. A laser emits photons of wavelength 680 nm. Find the energy per photon. Use $h = 6.626 \times 10^{-34}$ J s and $c = 3.00 \times 10^{8}$ m/s.

Answer

$E = hc/\lambda = (6.626 \times 10^{-34})(3.00 \times 10^{8})/(680 \times 10^{-9}) = 2.92 \times 10^{-19}$ J.

P18. A laser pulse contains 0.006 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

Answer

$N = E_{pulse}/E_{photon} = 0.006/(3.00 \times 10^{-19}) = 2 \times 10^{16}$ photons.

P19. A laser pulse contains 0.008 J. Each photon has energy 3.00×10^{-19} J. Find the number of photons.

Answer

$N = E_{pulse}/E_{photon} = 0.008/(3.00 \times 10^{-19}) = 2.67 \times 10^{16}$ photons.

P20. In a standing-wave picture, a circular path contains 4 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

Answer

$2\pi r = n\lambda$; circumference $= 4 \times 0.200 = 0.8$ nm.

P21. In a standing-wave picture, a circular path contains 5 complete de Broglie wavelengths of 0.200 nm each. Find the path circumference.

Answer

$2\pi r = n\lambda$; circumference $= 5 \times 0.200 = 1$ nm.

P22. Find the energy of a photon of frequency 3×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$E = hf = (6.626 \times 10^{-34})(3 \times 10^{14}) = 1.99 \times 10^{-19}$ J.

P23. Find the energy of a photon of frequency 4×10^{14} Hz. Use $h = 6.626 \times 10^{-34}$ J s.

Answer

$E = hf = (6.626 \times 10^{-34})(4 \times 10^{14}) = 2.65 \times 10^{-19}$ J.

P24. An atomic transition emits a photon of energy 3 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

Answer

λ = hc/E = 1240/3 = 413.3 nm.

P25. An atomic transition emits a photon of energy 3.5 eV. Calculate its wavelength in nanometres using hc ≈ 1240 eV nm.

Answer

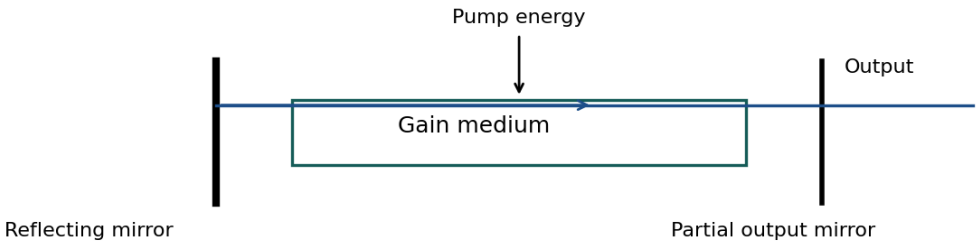
λ = hc/E = 1240/3.5 = 354.3 nm.

P26. Why does the quantum model replace exact Bohr paths with probability distributions?

Answer

Matter-wave behaviour and the uncertainty principle prevent a fully classical exact position-and-momentum trajectory from describing an electron state.

P27. Explain the role of mirrors in a simple laser cavity.



Answer

They provide optical feedback through the gain medium, encouraging repeated stimulated emission. One mirror transmits part of the amplified light as the output beam.

P28. Compare the emitted photons in stimulated emission with a general spontaneous-emission event.

Answer

An ideal stimulated photon matches the triggering photon's frequency, phase, direction and polarization. Spontaneous emission has no fixed phase relationship with an external triggering beam.

P29. A learner says a laser creates energy. Correct the statement using an energy balance.

Answer

A laser converts supplied pump energy into emitted light and other forms such as heat. Its output energy cannot exceed total input plus any decrease in stored internal energy.

P30. Describe how the standing-wave idea helps explain allowed states qualitatively.

Answer

Only wave patterns meeting the relevant boundary conditions form stable states. In the simple circular illustration, an integer number of wavelengths fits around the path; the full quantum model uses wavefunctions and allowed states rather than literal circular orbits.